

MH2100 Calculus III

Midterm Practice 2 (Solutions)

AY 2026/2027, Semester 1

Quantitative Research Society @NTU

Practice material

Overview

- **Coverage:** curves and tangent lines; limits, continuity, partial derivatives and total differentiability; chain rule, directional derivatives, tangent lines, linear approximation and the Implicit Function Theorem; extrema and Lagrange multipliers; order reversal, polar coordinates and volume by a height double integral.
- **Practice design:** four original problems, 100 marks in total; intended for 110 minutes. This practice design is not a confirmed 2026 examination format.
- **Current administration:** Tuesday 6 October 2026, 16:30–18:20, venue TBA. One double-sided typed A4 sheet is permitted; no attachments or sticky notes are permitted. No calculator policy is stated here.

Instructions

Answer **all** questions. Show all working clearly; answers without justification may receive zero marks.

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Question 1**(25 Marks)****PRACTICE PROBLEM 1.**

- (a) Let $\mathbf{r}(t) = (e^t, \sin t, t^2)$. Find the point at $t = 0$, a unit tangent vector there, and an equation of the tangent line. **(8 marks)**
- (b) Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $f(0, 0) = 0$ and $f(x, y) = x^2y/(x^2 + y^2)$ otherwise. Determine continuity at the origin, compute both coordinate partial derivatives there, and decide total differentiability there. Justify each conclusion. **(17 marks)**

Solution**Method 1: Direct calculation**

(a)

$$\mathbf{r}(0) = (1, 0, 0)$$

$$\mathbf{r}'(t) = (e^t, \cos t, 2t) \implies \mathbf{r}'(0) = (1, 1, 0), \quad |\mathbf{r}'(0)| = \sqrt{2}$$

$$\boxed{\mathbf{T}(0) = \frac{1}{\sqrt{2}}(1, 1, 0), \quad (x, y, z) = (1, 0, 0) + s(1, 1, 0)}$$

(b) $(x, y) \neq (0, 0)$:

$$|f(x, y)| = \frac{x^2|y|}{x^2 + y^2} \leq |y| \leq \sqrt{x^2 + y^2} \rightarrow 0 = f(0, 0) \quad (\text{Squeeze})$$

$$\implies f \text{ continuous at } (0, 0)$$

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0)}{h} = 0$$

$$f_y(0, 0) = \lim_{k \rightarrow 0} \frac{f(0, k)}{k} = 0$$

$$\text{if differentiable} \implies Df(0, 0) = (f_x, f_y) = (0, 0) \implies \frac{f(h, k)}{\sqrt{h^2 + k^2}} \rightarrow 0$$

$$\frac{f(t, t)}{\sqrt{2}t^2} = \frac{1}{2\sqrt{2}} \not\rightarrow 0 \quad (t \rightarrow 0^+) \quad (\text{contradiction})$$

$$\boxed{f \text{ is continuous but not totally differentiable at } (0, 0)}$$

UNOFFICIAL QRS marking scheme (25 marks)(a) **(8 marks)**(i) **(2 marks)** point;(ii) **(3 marks)** tangent vector and normalization;(iii) **(3 marks)** tangent line.(b) **(17 marks)**(i) **(5 marks)** radial bound proving continuity;(ii) **(4 marks)** partial derivatives;(iii) **(5 marks)** differentiability remainder;(iv) **(3 marks)** conclusions.

Question 2**(25 Marks)****PRACTICE PROBLEM 2.**

- (a) Let h be C^2 near $(0, 1)$, with $h(0, 1) = 2$, $\nabla h(0, 1) = (3, 2)$, and $h_{uu}(0, 1) = 2$, $h_{uv}(0, 1) = 1$, $h_{vv}(0, 1) = -3$. Define $G(x, y) = h(x - y, xy)$. At $P = (1, 1)$, find $G_{xy}(P)$, the directional derivative in the unit direction from P to $(3, 0)$, the tangent line to $G = 2$, and the linear approximation of $G(1.01, 0.98)$. **(15 marks)**
- (b) Let $F(x, y, z) = x + y^2 + e^z - 2$. Use the IFT near $(0, 1, 0)$ to write $z = q(x, y)$. Find $q_x(0, 1)$ and $q_y(0, 1)$, then linearly approximate $q(0.03, 0.99)$. **(10 marks)**

Solution**Method 1: Chain rule and implicit differentiation**

- (a) $u = x - y$, $v = xy$; at P : $(u, v) = (0, 1)$, $(u_x, u_y) = (1, -1)$, $(v_x, v_y) = (1, 1)$.

$$\nabla G = (h_u u_x + h_v v_x, h_u u_y + h_v v_y) \quad (\text{chain rule})$$

$$\nabla G(P) = (3 + 2, -3 + 2) = (5, -1)$$

$$\mathbf{u} = \frac{1}{\sqrt{5}}(2, -1)$$

$$D_{\mathbf{u}}G(P) = \frac{1}{\sqrt{5}}(5, -1) \cdot (2, -1) = \frac{11}{\sqrt{5}}$$

$$5(x - 1) - (y - 1) = 0$$

(tangent line to $G = 2$)

$$L(x, y) = 2 + 5(x - 1) - (y - 1)$$

$$G(1.01, 0.98) \approx L(1.01, 0.98) = 2.07$$

$$G_x = h_u + y h_v$$

$$G_{xy} = h_{uu}u_y + h_{uv}v_y + y(h_{vu}u_y + h_{vv}v_y) + h_v \quad (\text{Clairaut})$$

$$G_{xy}(P) = 2(-1) + 1(1) + (1(-1) + (-3)(1)) + 2 = -3$$

- (b) $F \in C^1$, $F(0, 1, 0) = 0$, $F_z(0, 1, 0) = 1 \neq 0 \implies$ IFT: $z = q(x, y)$ near $(0, 1)$.

$$q_x = -\frac{F_x}{F_z} = -1, \quad q_y = -\frac{F_y}{F_z} = -2 \quad \text{at } (0, 1)$$

$$q(0.03, 0.99) \approx 0 - 0.03 - 2(-0.01) = -0.01$$

UNOFFICIAL QRS marking scheme (25 marks)

- (a) **(15 marks)**
- (i) **(4 marks)** chain-rule gradient;
 - (ii) **(3 marks)** unit direction and derivative;
 - (iii) **(2 marks)** tangent line;
 - (iv) **(2 marks)** approximation;
 - (v) **(4 marks)** mixed second derivative.
- (b) **(10 marks)**
- (i) **(3 marks)** IFT hypotheses;

- (ii) **(4 marks)** derivatives;
- (iii) **(3 marks)** approximation.

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Question 3**(25 Marks)****PRACTICE PROBLEM 3.** Let $f(x, y) = x^2 - y^2$ on $C = \{(x, y) \in \mathbb{R}^2 : x^2 + 9y^2 \leq 9\}$.

- (a) Find and classify every interior critical point. **(8 marks)**
- (b) Use Lagrange multipliers to find every extremum candidate on $x^2 + 9y^2 = 9$. **(12 marks)**
- (c) Use the Extreme Value Theorem to state all absolute extrema on C . **(5 marks)**

Solution**Method 1: Lagrange multipliers**

(a)

$$\begin{aligned}\nabla f &= (2x, -2y) = \mathbf{0} \iff (x, y) = (0, 0) \\ D &= f_{xx}f_{yy} - f_{xy}^2 = 2(-2) - 0 = -4 < 0\end{aligned}\quad \text{(Second Derivative Test)}$$

 $(0, 0)$ is a saddle point

- (b) $g = x^2 + 9y^2$ is C^1 with $\nabla g = (2x, 18y) \neq \mathbf{0}$ on $g = 9 \implies$ Lagrange condition $\nabla f = \lambda \nabla g$ is necessary.

$$\begin{aligned}2x &= 2\lambda x \\ -2y &= 18\lambda y \\ x^2 + 9y^2 &= 9 \\ y = 0 &\implies x = \pm 3, f = 9 \\ x = 0 &\implies y = \pm 1, f = -1 \\ xy \neq 0 &\implies \lambda = 1 \text{ and } \lambda = -\frac{1}{9} \text{ impossible}\end{aligned}$$

 $(\pm 3, 0)$ with $f = 9$, $(0, \pm 1)$ with $f = -1$

- (c) f continuous, C compact $\implies$ extrema attained (Extreme Value Theorem); candidates: $f(0, 0) = 0$ and the values in (b).

$\max_C f = 9 \text{ at } (\pm 3, 0), \quad \min_C f = -1 \text{ at } (0, \pm 1)$

Method 2: direct bounds

$$\begin{aligned}x^2 \leq 9 - 9y^2, y^2 \geq 0 &\implies f \leq 9 - 10y^2 \leq 9 && \text{(equality: } y = 0, x = \pm 3) \\ x^2 \geq 0, y^2 \leq 1 &\implies f \geq -y^2 \geq -1 && \text{(equality: } x = 0, y = \pm 1)\end{aligned}$$

UNOFFICIAL QRS marking scheme (25 marks)(a) **(8 marks)**

- (i) **(3 marks)** critical point;
- (ii) **(3 marks)** discriminant;
- (iii) **(2 marks)** classification.

(b) **(12 marks)**

- (i) **(4 marks)** Lagrange system;
 - (ii) **(4 marks)** all branches;
 - (iii) **(4 marks)** candidates and values.
- (c) **(5 marks)**
- (i) **(2 marks)** EVT hypotheses;
 - (ii) **(3 marks)** global extrema.

Question 4**(25 Marks)****PRACTICE PROBLEM 4.**

- (a) Reverse the order of integration and evaluate

$$\int_0^1 \left[\int_{\sqrt{y}}^1 (1+x) dx \right] dy.$$

(10 marks)

- (b) Let
- D
- be the part of
- $x^2 + y^2 \leq 4y$
- in the first quadrant. Find the volume above
- D
- and below
- $z = 20 - x^2 - y^2$
- , using polar coordinates and a height double integral.
- (15 marks)**

Solution**Method 1: Reversal and polar height integral**

- (a) Region
- $\{0 \leq y \leq 1, \sqrt{y} \leq x \leq 1\} = \{0 \leq x \leq 1, 0 \leq y \leq x^2\}$
- .

$$\begin{aligned} \int_0^1 \int_{\sqrt{y}}^1 (1+x) dx dy &= \int_0^1 \int_0^{x^2} (1+x) dy dx && \text{(Fubini)} \\ &= \boxed{\int_0^1 (x^2 + x^3) dx = \frac{7}{12}} \end{aligned}$$

- (b)
- D
- :
- $r^2 \leq 4r \sin \theta$
- , i.e.
- $0 \leq \theta \leq \frac{\pi}{2}$
- ,
- $0 \leq r \leq 4 \sin \theta$
- ;
- $20 - r^2 \geq 4 > 0$
- on
- D
- .

$$\begin{aligned} V &= \int_0^{\pi/2} \int_0^{4 \sin \theta} (20 - r^2) r dr d\theta && (dA = r dr d\theta) \\ &= \int_0^{\pi/2} (160 \sin^2 \theta - 64 \sin^4 \theta) d\theta \\ &= 160 \cdot \frac{\pi}{4} - 64 \cdot \frac{3\pi}{16} \\ &= \boxed{28\pi} \end{aligned}$$

UNOFFICIAL QRS marking scheme (25 marks)

- (a)
- (10 marks)**

- (i) **(3 marks)** reversed region;
- (ii) **(3 marks)** bounds;
- (iii) **(2 marks)** inner integral;
- (iv) **(2 marks)** value $7/12$.

- (b)
- (15 marks)**

- (i) **(4 marks)** polar geometric reasoning;
- (ii) **(3 marks)** height and Jacobian;
- (iii) **(5 marks)** double integral;
- (iv) **(3 marks)** value 28π . Equivalent correct Cartesian work receives full credit.